Question I

Let A be an $m \times n$ matrix and its singular value decomposition (SVD) be

$$V^T A U = F = \begin{bmatrix} \mu_1 & 0 & \cdots & 0 & \cdots & 0 \\ 0 & \mu_2 & \cdots & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & \mu_r & \cdots & 0 \\ 0 & 0 & \cdots & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & 0 & \cdots & 0 \end{bmatrix},$$

where the matrices U and V are orthogonal, and the singular values μ_i satisfy

$$\mu_1 \geq \mu_2 \geq \cdots \geq \mu_r > 0.$$

Define the $n \times m$ matrix

$$F^+ = \begin{bmatrix} \mu_1^{-1} & 0 & \cdots & 0 & \cdots & 0 \\ 0 & \mu_2^{-1} & \cdots & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & \mu_r^{-1} & \cdots & 0 \\ 0 & 0 & \cdots & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & 0 & \cdots & 0 \end{bmatrix}$$

and

$$A^+ = U\, F^+\, V^T.$$

The matrix A^+ is called as the *generalized inverse* of A.

1. Show that $x^* = A^+ b$ is the least square solution of the linear system $Ax = b$.
2. Show that
$$\lim_{\alpha \to 0^+} (\alpha I + A^T A)^{-1} A^T = A^+,$$
where $\alpha > 0$.

Question II

Consider the following finite difference scheme

$$\frac{v_m^{n+1} - v_m^{n-1}}{2\tau} + a\Big(1 + \frac{h^2}{6}\delta^2\Big)^{-1} \delta_0 v_m^n = f_m^n$$

for the transport equation $u_t + a u_x = f$, where a is a constant, $\tau > 0$ is the time step, δ_0 denotes the standard second order central difference operator for u_x, and δ^2

denotes the standard second order central difference operator for u_{xx}. Assume that $\lambda = \tau/h$ is a constant with h being the spatial mesh size. Show that the scheme is stable if and only if

$$|a\lambda| < \frac{1}{\sqrt{3}}.$$

How do you modify the scheme when a is not a constant without affecting the accuracy of the scheme? Justify your answer.

Question III

Show that the following system of ordinary differential equations

$$\begin{aligned} \frac{dx}{dt} &= 0.5x + 2.5y - x(x^2 + y^2), \\ \frac{dy}{dt} &= -0.5x + 1.5y - y(x^2 + y^2). \end{aligned}$$

has at least one periodic solution.